Industrial Component Inspection System

Two cameras. One Jetson Orin. A custom AI that keeps learning new parts, and only deploys what passes validation.



System architecture

Edge sees, core decides and remembers, access shows and signals

EDGE: SEES

2 RTSP cameras
192.168.1.10 / .20

PoE + router
network and power

Jetson Orin
192.168.1.100, live vision

CORE: DECIDES

FastAPI backend
BOM validation, APIs

SQL Server
records, users, history

File + model store
images, masks, models

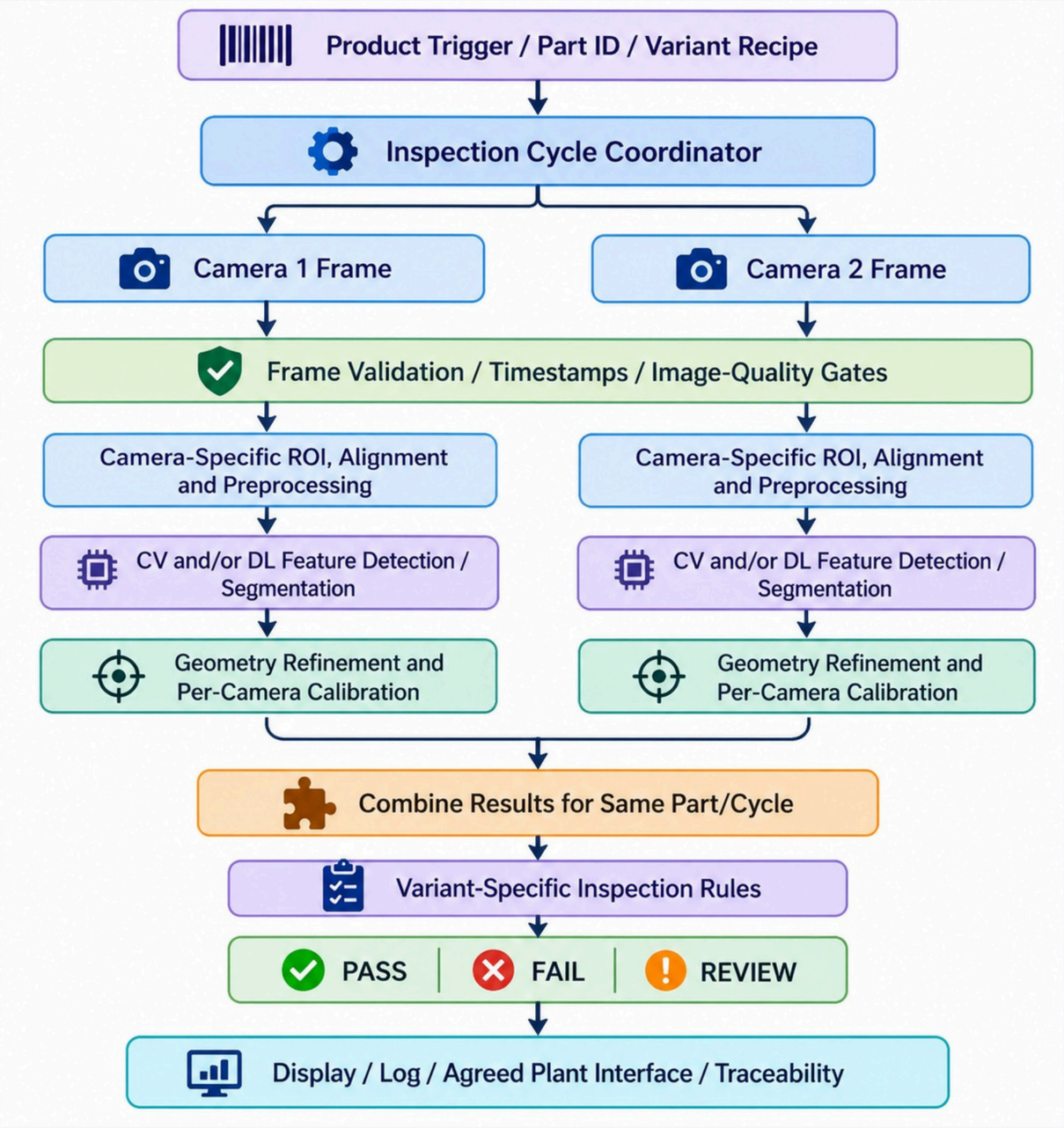
ACCESS: SHOWS

React + Vite dashboard
live, results, components

Secure remote access
TLS , roles, audit

PLC
PASS / ERROR / IN PROGRESS

System architecture



Inspect in real time. Learn offline. Never deploy an unvalidated model

INSPECT | live on the Jetson

1. Two streams, newest frames only
2. Green product outline
3. Colored masks + labels
4. Track + merge both cameras
5. Count vs BOM: PASS / ERROR

LEARN | offline on the server

1. Register component + images
2. Reference match (fast path)
3. Train candidate model
4. Pass the 7 validation gates
5. Deploy, health check, activate

7 validation gates before a model goes live

1 count per physical part, across both cameras

3 outcomes: PASS, ERROR, INCOMPLETE

Execution plan: eight phases

Foundation first, intelligence next, delivery last

FOUNDATION | **INTELLIGENCE** | **DELIVERY**

1	2	3	4	5	6	7	8
Audit + preserve	**Management server**	**Reference recognition**	**Detection**	**Automated learning**	**Deploy**	**Counting + BOM**	**Remote + PLC**
Cameras, code, dataset, runtime	Auth, upload, SQL, storage	Encoder vs similar parts	Component Identification and Dimensional Analysis	Datasets, training, gates	Jetson, health	Dedupe, dashboard, history	Secure access, PLC last
A B	B	A	A	A	A	A B	B

Owner: **A** Vision + AI **B** Platform

Camera 1 file + autostart untouched

Model live only after all gates pass

PLC only once workflow is reliable

Division of work: two owners

Person A: Vision + AI

- Cameras + Jetson capture
- FastAPI + API groups
- Reference recognition
- Auth, roles, audit logs
- Tracking + 2-camera merge
- Dashboard + history
- Model packaging
- TLS, backups

Agreed contracts

Person B: Platform

- Product outline + masks
- SQL Server + file storage
- Custom model
- Component portal + upload
- Training + validation tests
- BOM validation + records
- Jetson deploy
- Optional PLC signaling

TOGETHER

Dataset audit | Interface contracts: inspection result with model version, model + class-map package, camera status | Validation gate thresholds | Count accuracy vs manual checks | Go-live review